Laboratorio 3

Planificación IP, VLSM, VLANs y enrutamiento OSPF / EIGRP con redistribución

PDF 2: evidencias de configuración y conectividad

Redes - Sección 10
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1. Alcance

Este documento reúne las evidencias pedidas en el laboratorio: las tablas de enrutamiento de los cinco routers, la demostración del balanceo de costo igual en OSPF y del balanceo 75/25 en EIGRP, la redistribución mutua con rutas resumidas en R-CENTRAL y las pruebas de conectividad extremo a extremo.

2. Topología y direccionamiento

Cinco routers conectados por seis enlaces seriales forman dos triángulos que comparten a R-CENTRAL: el lado OSPF (área 0) con R-OSPF-1 y R-OSPF-2, y el lado EIGRP (AS 100) con R-EIGRP-1 y R-EIGRP-2. R-OSPF-1 y R-EIGRP-1 llevan VLANs con router-on-a-stick; R-OSPF-2 y R-EIGRP-2 tienen una LAN simple. R-CENTRAL no tiene LAN propia.

Enlaces seriales

Enlace	Extremo A	Extremo B	Red
OSPF-1 / OSPF-2	R-OSPF-1 Se0/0/0 (10.0.0.1)	R-OSPF-2 Se0/0/0 (10.0.0.2)	10.0.0.0/30
OSPF-1 / CENTRAL	R-OSPF-1 Se0/0/1 (10.0.0.5)	R-CENTRAL Se0/0/0 (10.0.0.6)	10.0.0.4/30
OSPF-2 / CENTRAL	R-OSPF-2 Se0/0/1 (10.0.0.9)	R-CENTRAL Se0/0/1 (10.0.0.10)	10.0.0.8/30
EIGRP-1 / EIGRP-2	R-EIGRP-1 Se0/0/0 (10.0.0.13)	R-EIGRP-2 Se0/0/0 (10.0.0.14)	10.0.0.12/30
EIGRP-1 / CENTRAL	R-EIGRP-1 Se0/0/1 (10.0.0.17)	R-CENTRAL Se0/1/0 (10.0.0.18)	10.0.0.16/30
EIGRP-2 / CENTRAL	R-EIGRP-2 Se0/0/1 (10.0.0.21)	R-CENTRAL Se0/1/1 (10.0.0.22)	10.0.0.20/30

Redes de usuario

Router	Segmento	Red	Gateway
R-OSPF-1	VLAN 10 datos	192.168.1.0/26	192.168.1.1
R-OSPF-1	VLAN 30 administración	192.168.1.64/28	192.168.1.65
R-OSPF-2	LAN datos	192.168.0.0/24	192.168.0.1
R-EIGRP-1	VLAN 10 datos	172.16.0.128/26	172.16.0.129
R-EIGRP-1	VLAN 20 voz	172.16.0.192/27	172.16.0.193
R-EIGRP-1	VLAN 30	172.16.0.224/28	172.16.0.225
R-EIGRP-2	LAN datos	172.16.0.0/25	172.16.0.1

El enunciado indica tres VLANs para R-EIGRP-1 pero solo detalla dos; la tercera (VLAN 30) se agregó en el siguiente bloque libre, un /28.

3. Tablas de enrutamiento (show ip route)

Los códigos que aparecen son C (conectada), L (local), O (OSPF interna), O E1 (OSPF externa tipo 1), D (EIGRP interna) y S (estática).

3.1 R-OSPF-1

```
01:20:38: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/0/1 from LOADING to FULL, Loading


R-OSPF-1>
R-OSPF-1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/20] via 10.0.0.6, 00:01:24, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/30] via 10.0.0.6, 00:01:24, Serial0/0/1
O    192.168.0.0/24 [110/21] via 10.0.0.2, 00:01:24, Serial0/0/0
                    [110/21] via 10.0.0.6, 00:01:24, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30

R-OSPF-1>
R-OSPF-1>
R-OSPF-1>
```

R-OSPF-1: las VLAN locales aparecen conectadas; la LAN de R-OSPF-2 (192.168.0.0/24) tiene dos rutas OSPF de igual métrica; todo el lado EIGRP llega como una sola ruta externa O E1 172.16.0.0/24.

3.2 R-OSPF-2

```
R-OSPF-2>
R-OSPF-2>
R-OSPF-2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

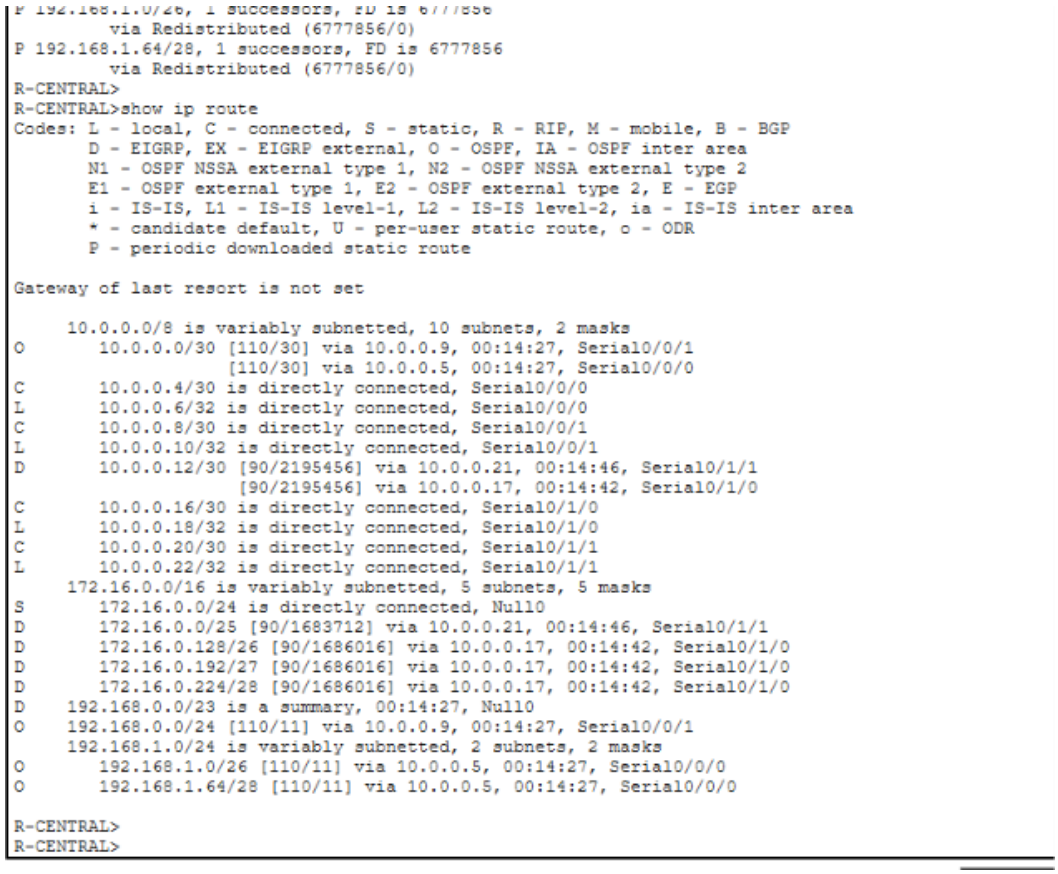
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/20] via 10.0.0.10, 00:18:03, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/30] via 10.0.0.10, 00:18:03, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/21] via 10.0.0.1, 00:18:03, Serial0/0/0
                       [110/21] via 10.0.0.10, 00:18:03, Serial0/0/1
O       192.168.1.64/28 [110/21] via 10.0.0.1, 00:18:03, Serial0/0/0
                        [110/21] via 10.0.0.10, 00:18:03, Serial0/0/1

R-OSPF-2>
R-OSPF-2>
R-OSPF-2>
```

R-OSPF-2: las dos VLAN de R-OSPF-1 (192.168.1.0/26 y 192.168.1.64/28) se alcanzan por dos next-hops de métrica 21; el dominio EIGRP aparece como la externa O E1 172.16.0.0/24.

3.3 R-CENTRAL



```
P 192.168.1.0/26, 1 successors, FD is 6777856
         via Redistributed (6777856/0)
P 192.168.1.64/28, 1 successors, FD is 6777856
         via Redistributed (6777856/0)
R-CENTRAL>
R-CENTRAL>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/30] via 10.0.0.9, 00:14:27, Serial0/0/1
                    [110/30] via 10.0.0.5, 00:14:27, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
D       10.0.0.12/30 [90/2195456] via 10.0.0.21, 00:14:46, Serial0/1/1
                     [90/2195456] via 10.0.0.17, 00:14:42, Serial0/1/0
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.18/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.22/32 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/1683712] via 10.0.0.21, 00:14:46, Serial0/1/1
D       172.16.0.128/26 [90/1686016] via 10.0.0.17, 00:14:42, Serial0/1/0
D       172.16.0.192/27 [90/1686016] via 10.0.0.17, 00:14:42, Serial0/1/0
D       172.16.0.224/28 [90/1686016] via 10.0.0.17, 00:14:42, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:14:27, Null0
O    192.168.0.0/24 [110/11] via 10.0.0.9, 00:14:27, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/11] via 10.0.0.5, 00:14:27, Serial0/0/0
O       192.168.1.64/28 [110/11] via 10.0.0.5, 00:14:27, Serial0/0/0

R-CENTRAL>
R-CENTRAL>
```

R-CENTRAL: mantiene las rutas específicas de ambos dominios (O y D) y genera los dos resúmenes: 192.168.0.0/23 hacia EIGRP (marcado como summary a Null0) y 172.16.0.0/24 hacia OSPF (ruta estática a Null0).

3.4 R-EIGRP-1

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.14 (Serial0/0/0) is up: new adjacency

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.18 (Serial0/0/1) is up: new adjacency


R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/11604480] via 10.0.0.18, 00:09:31, Serial0/0/1
D EX    10.0.0.4/30 [170/11604480] via 10.0.0.18, 00:09:46, Serial0/0/1
D EX    10.0.0.8/30 [170/11604480] via 10.0.0.18, 00:09:46, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.13/32 is directly connected, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/1
L       10.0.0.17/32 is directly connected, Serial0/0/1
D       10.0.0.20/30 [90/6510080] via 10.0.0.18, 00:09:46, Serial0/0/1
                     [90/6996480] via 10.0.0.14, 00:09:49, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 7 subnets, 5 masks
D       172.16.0.0/25 [90/2170112] via 10.0.0.14, 00:09:49, Serial0/0/0
                      [90/6510336] via 10.0.0.18, 00:09:46, Serial0/0/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/27 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
C       172.16.0.224/28 is directly connected, GigabitEthernet0/0.30
L       172.16.0.225/32 is directly connected, GigabitEthernet0/0.30
D    192.168.0.0/23 [90/6996480] via 10.0.0.18, 00:09:31, Serial0/0/1

R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>
```

R-EIGRP-1: tiene conectadas sus tres VLAN, alcanza la LAN de R-EIGRP-2 (172.16.0.0/25) por dos caminos y recibe todo el dominio OSPF como un solo prefijo D 192.168.0.0/23.

3.5 R-EIGRP-2

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.13 (Serial0/0/0) is up: new adjacency


R-EIGRP-2>
R-EIGRP-2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/11604480] via 10.0.0.22, 00:09:19, Serial0/0/1
D EX    10.0.0.4/30 [170/11604480] via 10.0.0.22, 00:09:36, Serial0/0/1
D EX    10.0.0.8/30 [170/11604480] via 10.0.0.22, 00:09:38, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.14/32 is directly connected, Serial0/0/0
D       10.0.0.16/30 [90/6510080] via 10.0.0.22, 00:09:34, Serial0/0/1
                     [90/6996480] via 10.0.0.13, 00:09:37, Serial0/0/0
C       10.0.0.20/30 is directly connected, Serial0/0/1
L       10.0.0.21/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/2172416] via 10.0.0.13, 00:09:37, Serial0/0/0
                        [90/6512640] via 10.0.0.22, 00:09:34, Serial0/0/1
D       172.16.0.192/27 [90/2172416] via 10.0.0.13, 00:09:37, Serial0/0/0
                        [90/6512640] via 10.0.0.22, 00:09:34, Serial0/0/1
D       172.16.0.224/28 [90/2172416] via 10.0.0.13, 00:09:37, Serial0/0/0
                        [90/6512640] via 10.0.0.22, 00:09:34, Serial0/0/1
D    192.168.0.0/23 [90/6996480] via 10.0.0.22, 00:09:19, Serial0/0/1

R-EIGRP-2>
R-EIGRP-2>
R-EIGRP-2>
```

R-EIGRP-2: las tres VLAN de R-EIGRP-1 tienen un camino directo y uno alterno por R-CENTRAL; el dominio OSPF llega resumido en 192.168.0.0/23.

4. Balanceo OSPF de costo igual

```
R-OSPF-1>show ip ospf interface brief
Interface     PID   Area                    IP Address/Mask          Cost  State  Nbrs F/C
Gig             1   0                 192.168.1.1/255.255.255.192   1       DR  0/0
Gig             1   0                192.168.1.65/255.255.255.240   1       DR  0/0
Se0/0/1         1   0                    10.0.0.5/255.255.255.252   10   POINT  0/0
Se0/0/0         1   0                    10.0.0.1/255.255.255.252   20   POINT  0/0
```

R-OSPF-1, show ip ospf interface brief: Serial0/0/0 (enlace directo hacia R-OSPF-2) cuesta 20 y Serial0/0/1 (hacia R-CENTRAL) cuesta 10.

El camino directo hacia R-OSPF-2 cuesta 20. El camino por R-CENTRAL cuesta 10 + 10 = 20. Al ser iguales, OSPF instala los dos caminos y reparte el tráfico entre ellos, lo que se confirma en las secciones 3.1 y 3.2, donde las redes de la otra sede aparecen con dos next-hops y la misma métrica.

5. Balanceo EIGRP 75/25

```
R-EIGRP-1>show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(4.4.4.4)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 11604480
         via Rstatic (11604480/6777856)
P 10.0.0.4/30, 1 successors, FD is 11604480
         via Rstatic (11604480/6777856)
P 10.0.0.8/30, 1 successors, FD is 11604480
         via Rstatic (11604480/6777856)
P 10.0.0.12/30, 1 successors, FD is 2169856
         via Connected, Serial0/0/0
P 10.0.0.16/30, 1 successors, FD is 6484480
         via Connected, Serial0/0/1
P 10.0.0.20/30, 2 successors, FD is 6510080
         via 10.0.0.18 (6510080/1683456), Serial0/0/1
         via 10.0.0.14 (6996480/6484480), Serial0/0/0
P 172.16.0.0/25, 2 successors, FD is 2170112
         via 10.0.0.14 (2170112/2816), Serial0/0/0
         via 10.0.0.18 (6510336/1683712), Serial0/0/1
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.10
P 172.16.0.192/27, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.20
P 172.16.0.224/28, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.30
P 192.168.0.0/23, 1 successors, FD is 6996480
         via 10.0.0.18 (6996480/2169856), Serial0/0/1
R-EIGRP-1>
```

R-EIGRP-1, show ip eigrp topology: la red 172.16.0.0/25 tiene dos rutas instaladas. Directa por 10.0.0.14: métrica 2170112 (RD 2816). Alterna por 10.0.0.18: métrica 6510336 (RD 1683712).

Camino	Siguiente salto	Métrica (FD)	Distancia reportada (RD)	Papel
Directo	10.0.0.14 (Se0/0/0)	2.170.112	2.816	Successor
Por R-CENTRAL	10.0.0.18 (Se0/0/1)	6.510.336	1.683.712	Feasible successor

La ruta alterna es factible porque su distancia reportada (1.683.712) es menor que la FD de la principal (2.170.112). Además su métrica es exactamente 3 veces la principal (6.510.336 / 2.170.112 = 3), así que con **variance 3** EIGRP la instala también y reparte el tráfico de forma inversamente proporcional: 3 partes por la ruta directa y 1 por la alterna, es decir, 75% y 25%.

```
R-EIGRP-1>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 4.4.4.4
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 3

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     172.16.0.128/26
     172.16.0.192/27
     10.0.0.12/30
     10.0.0.16/30
     172.16.0.224/28
  Passive Interface(s):
    Vlan1
    GigabitEthernet0/0
    GigabitEthernet0/1
    GigabitEthernet0/0.10
    GigabitEthernet0/0.20
    GigabitEthernet0/0.30
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.14       90            6999
    10.0.0.18       90            10008
  Distance: internal 90 external 170

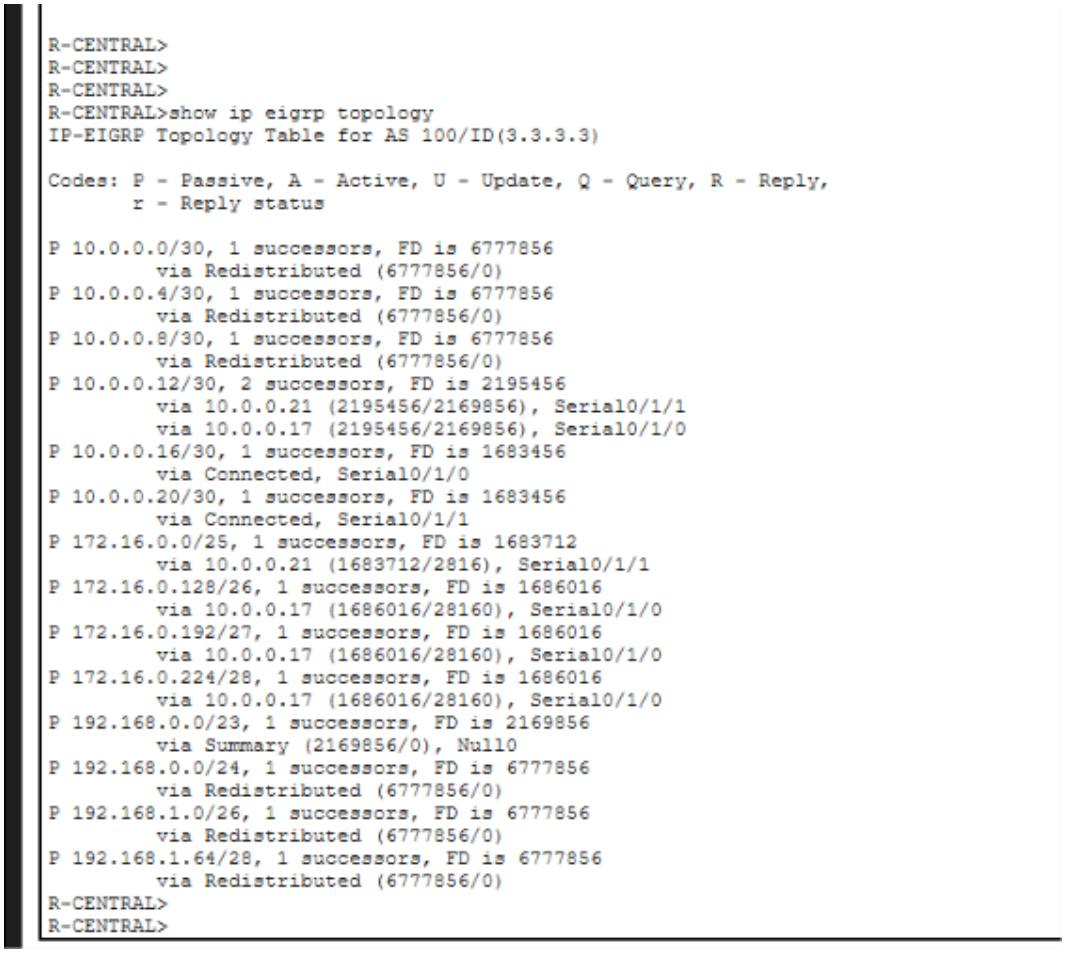
R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>show running-config
```

R-EIGRP-1, show ip protocols: AS 100, K1 = K3 = 1, router-id 4.4.4.4, variance 3 y las interfaces de VLAN configuradas como pasivas.

6. Redistribución y sumarización

R-CENTRAL es el único router que corre OSPF y EIGRP a la vez, y es donde se redistribuyen mutuamente las rutas.

6.1 OSPF hacia EIGRP



```
R-CENTRAL>
R-CENTRAL>
R-CENTRAL>
R-CENTRAL>show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(3.3.3.3)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 6777856
         via Redistributed (6777856/0)
P 10.0.0.4/30, 1 successors, FD is 6777856
         via Redistributed (6777856/0)
P 10.0.0.8/30, 1 successors, FD is 6777856
         via Redistributed (6777856/0)
P 10.0.0.12/30, 2 successors, FD is 2195456
         via 10.0.0.21 (2195456/2169856), Serial0/1/1
         via 10.0.0.17 (2195456/2169856), Serial0/1/0
P 10.0.0.16/30, 1 successors, FD is 1683456
         via Connected, Serial0/1/0
P 10.0.0.20/30, 1 successors, FD is 1683456
         via Connected, Serial0/1/1
P 172.16.0.0/25, 1 successors, FD is 1683712
         via 10.0.0.21 (1683712/2816), Serial0/1/1
P 172.16.0.128/26, 1 successors, FD is 1686016
         via 10.0.0.17 (1686016/28160), Serial0/1/0
P 172.16.0.192/27, 1 successors, FD is 1686016
         via 10.0.0.17 (1686016/28160), Serial0/1/0
P 172.16.0.224/28, 1 successors, FD is 1686016
         via 10.0.0.17 (1686016/28160), Serial0/1/0
P 192.168.0.0/23, 1 successors, FD is 2169856
         via Summary (2169856/0), Null0
P 192.168.0.0/24, 1 successors, FD is 6777856
         via Redistributed (6777856/0)
P 192.168.1.0/26, 1 successors, FD is 6777856
         via Redistributed (6777856/0)
P 192.168.1.64/28, 1 successors, FD is 6777856
         via Redistributed (6777856/0)
R-CENTRAL>
R-CENTRAL>
```

R-CENTRAL, show ip eigrp topology: las redes del lado OSPF entran a EIGRP como Redistributed; el resumen 192.168.0.0/23 aparece como Summary hacia Null0.

Los routers EIGRP reciben solo el resumen 192.168.0.0/23 (secciones 3.4 y 3.5), no cada subred por separado.

6.2 EIGRP hacia OSPF

R-CENTRAL anuncia a OSPF un único prefijo, 172.16.0.0/24, que los routers OSPF reciben como ruta externa tipo 1 (O E1, secciones 3.1 y 3.2). Las rutas específicas (/25, /26, /27 y /28) siguen presentes en R-CENTRAL y tienen prioridad sobre el /24 por prefijo más largo, de modo que la ruta a Null0 no descarta tráfico válido.

```
R-CENTRAL>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  down                  down
GigabitEthernet0/1     unassigned      YES unset  down                  down
Serial0/0/0            10.0.0.6        YES manual up                    up
Serial0/0/1            10.0.0.10       YES manual up                    up
Serial0/1/0            10.0.0.18       YES manual up                    up
Serial0/1/1            10.0.0.22       YES manual up                    up
Vlan1                  unassigned      YES unset  up                    down
R-CENTRAL>show ip ospf neighbor

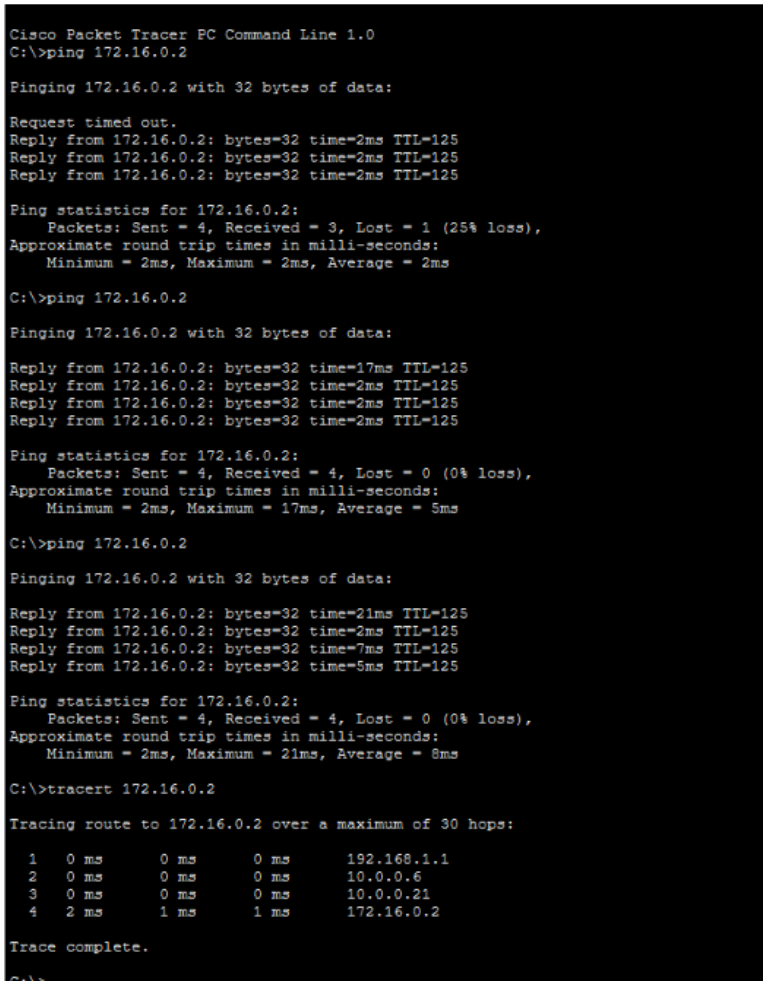

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:37    10.0.0.5        Serial0/0/0
2.2.2.2           0   FULL/  -        00:00:37    10.0.0.9        Serial0/0/1
R-CENTRAL>show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)       Cnt  Num
0   10.0.0.21       Se0/1/1        10   00:00:54  40     1000  0   22
1   10.0.0.17       Se0/1/0        14   00:00:50  40     1000  0   20

R-CENTRAL>
```

R-CENTRAL: vecinos OSPF 1.1.1.1 y 2.2.2.2 en FULL, vecinos EIGRP 10.0.0.21 y 10.0.0.17 en el proceso 100 y las cuatro interfaces seriales activas.

7. Conectividad extremo a extremo

Prueba hecha desde un PC de la VLAN 10 de R-OSPF-1 hacia el PC 172.16.0.2 de R-EIGRP-2. El primer ping pierde un paquete mientras se resuelve ARP; los siguientes responden 4 de 4 (0% de pérdida). El tracert llega en cuatro saltos.



Ping y tracert hacia 172.16.0.2. Ruta: 192.168.1.1 (R-OSPF-1), 10.0.0.6 (R-CENTRAL), 10.0.0.21 (R-EIGRP-2) y el PC destino.

8. Conclusiones

Los dos dominios se comunican de extremo a extremo a través de R-CENTRAL. OSPF conserva dos caminos de costo igual entre sus routers, EIGRP instala un successor y un feasible successor con variance 3 para repartir 75/25, y R-CENTRAL intercambia una sola ruta resumida por cada dominio, lo que mantiene pequeñas las tablas de enrutamiento.